

	1	2	3	4	5	6
A	STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2 MicroScout flight controller – root sheet. Sub-sheets: Power, MCU and USB, Sensors and expanders, Camera, LEDs, buzzer, receiver. Generated by tools/sch from hardware/*/design/*.py – edit the Python, not this file, until G3 hand-off. Connectivity: net labels and power symbols; check report in review/G2/.					A
B	Power File: power.kicad_sch	MCU and USB File: mcu.kicad_sch	Sensors and expanders File: sensors.kicad_sch	Camera, LEDs, buzzer, receiver File: io.kicad_sch		B
C						C
D	Generated schematic – not reviewed. Values are first-pass G2 choices. STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2 MicroScout (open hardware, CERN-OHL-S-2.0) Sheet: / File: microscout-fc.kicad_sch Title: MicroScout flight controller – root Size: A4 Date: 2026-10-08 Rev: G2-draft-1 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 1/5					D
	1	2	3	4	5	6

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

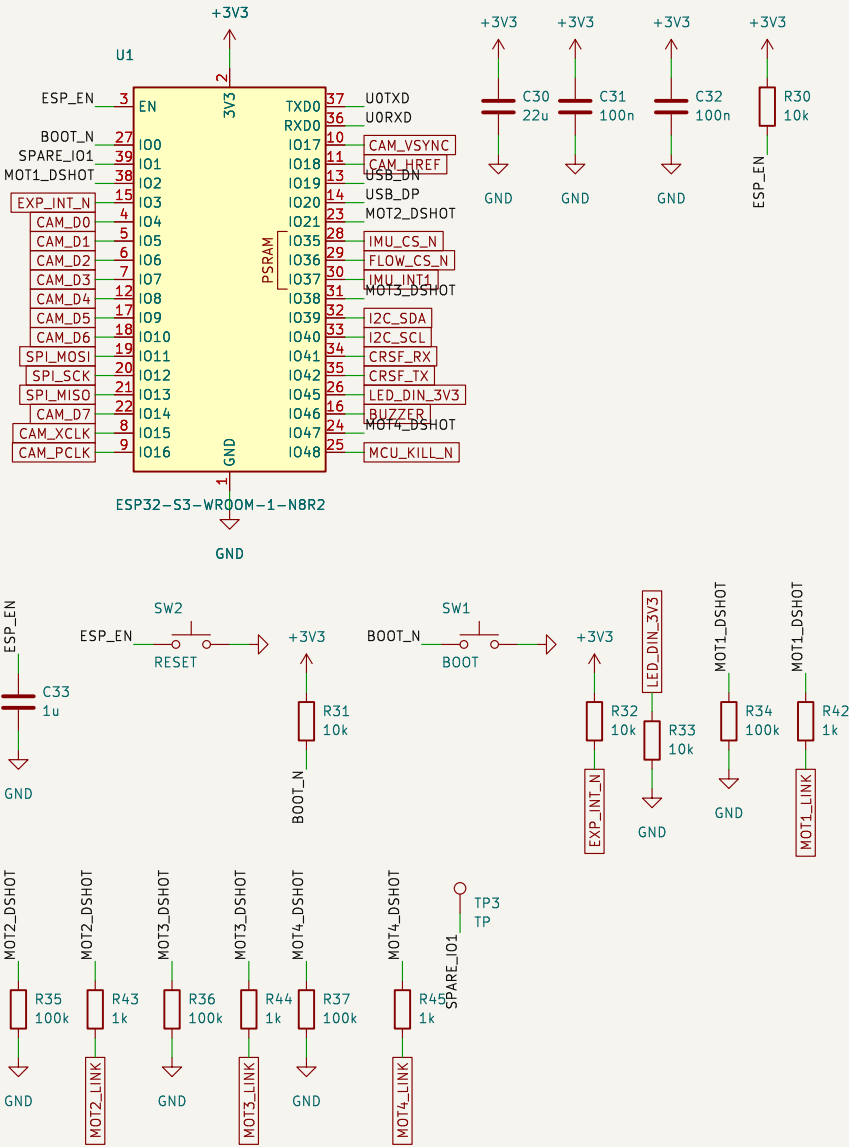
MicroScout flight controller – MCU and USB

ESP32-S3-WROOM-1-N8R2 (pin table: review/G1/pin-allocation.csv; module datasheet Table 3-1, section 9)

3V3: 22 uF + 2x 0.1 uF (datasheet peripheral schematic). EN: 10 k + 1 uF RC (datasheet section 9).

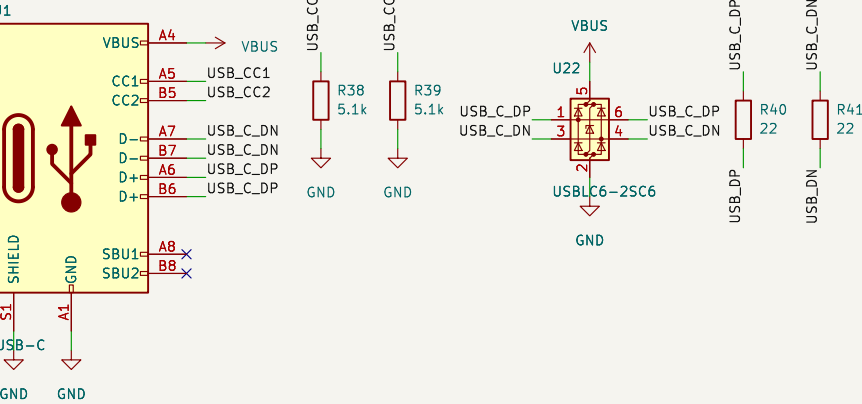
Strapping: GPIO0 10 k up (BOOT button to GND), GPIO3 10 k up, GPIO45 10 k down, GPIO46 100 k down (at Q3).

GPIO1 is spare (test point). DShot outputs have 100 k pull-downs (motor stopped during reset) and 1 k series resistors that limit current into an unpowered ESC MCU (e.g. USB-only power). GPIO48 reaches KILL through D5.



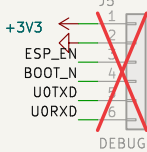
USB-C (device, USB 2.0) with ESD protection

CC1/CC2 5.1 k to GND (sink, Rd). USBLC6-2SC6 at the connector; 22 Ohm series resistors near the module (ESP32-S3 checklist 'initial value 22/33 Ohm'). VBUS feeds the charger and the logic OR-ing only.



Debug pads (UART0 ROM log, EN, IO0) – optional 1.27 mm header, not fitted

Normal flashing and logging use native USB (GPIO19/20).



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN-OHL-S-2.0)

Sheet: /MCU and USB/
File: mcu.kicad_sch

Title: MicroScout flight controller – MCU and USB

Size: A3 Date: 2026-10-08

Rev: G2-draft-1

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

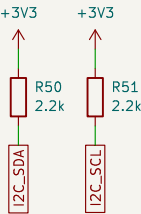
Id: 3/5

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout flight controller – Sensors and expanders

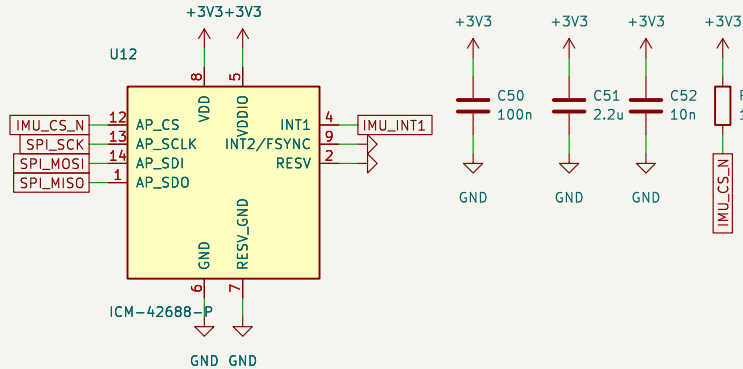
I2C bus (400 kHz) pull-ups

2.2 k: rise time 0.8473 x 2.2 k x C_{bus}; 300 ns limit -> C_{bus} <= 160 pF. Estimate ~150 pF with 4 ToF wires (G2 estimate, measure at G6; 1.8 k is the fallback). Devices: 0x20/0x21 expanders, 0x29 ToF (re-addressed), 0x2C mag, 0x30 camera SCCB, 0x40 INA226, 0x6B charger (datasheet also says 0x6A), 0x76 baro.
The Flow deck's VL53L1x sits at 0x29 with no XSHUT; firmware re-addresses it first, before releasing any satellite.



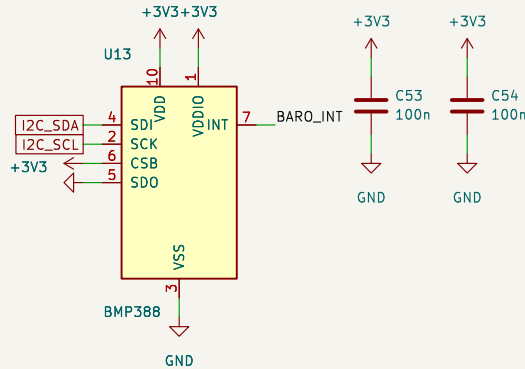
IMU – ICM-42688-P on SPI2 (datasheet DS-000347 Table 10/11, Fig. 7)

VDD 0.1 uF + 2.2 uF, VDDIO 10 nF, RESV pins to GND; INT2/FSYNC to GND (unused), 4-wire SPI.
CS pull-up 10 k keeps the IMU deselected while the ESP32 boots.



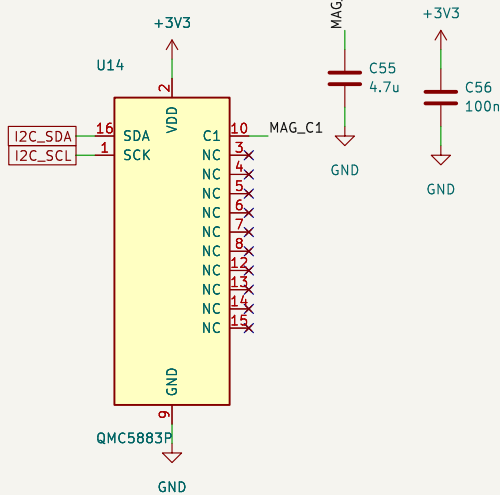
Barometer – BMP388 fitted (BMP390 same footprint; BMP390 out of stock 2026-10-07, OQ-6)

I2C: CSB to VDDIO, SDO to GND -> 0x76. 100 nF on VDD and VDDIO (datasheet Fig. 25).



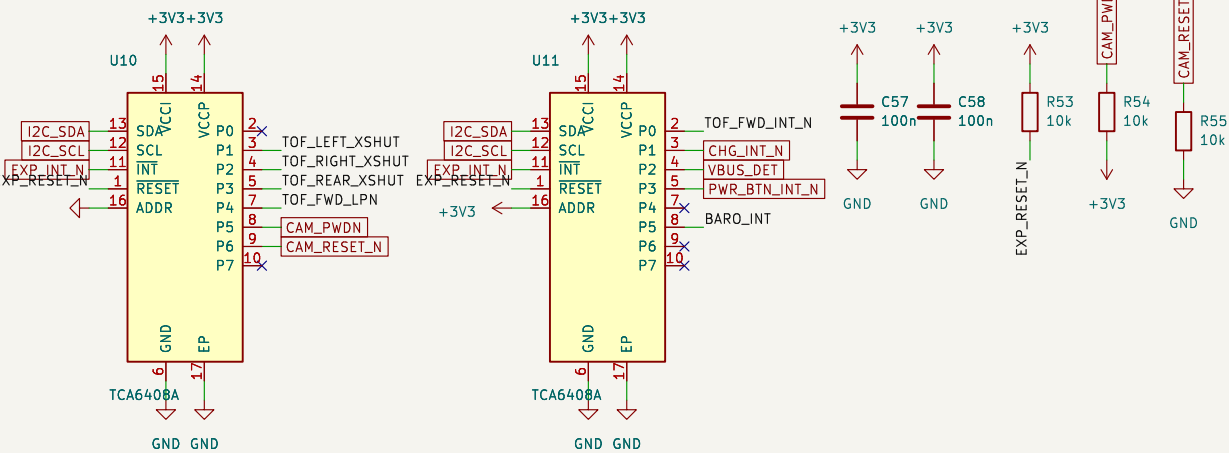
Magnetometer – QMC5883P (address 0x2C fixed)

C1 pin: 4.7 uF, ESR < 200 mOhm (datasheet 4.3.3). 100 nF VDD decoupling added by the agent (not in the datasheet).



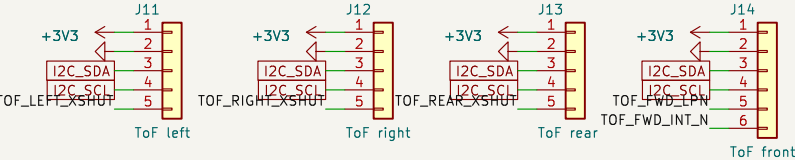
I2C GPIO expanders – TCA6408A x2 (0x20 outputs, 0x21 inputs); RESET pulled up, INT wired-OR to GPIO3

Expander pins power up as inputs, so the external pulls set the safe state: ToF in reset (XSHUT low, LPn low on the satellite), camera powered down (PWDN high) and held in reset.



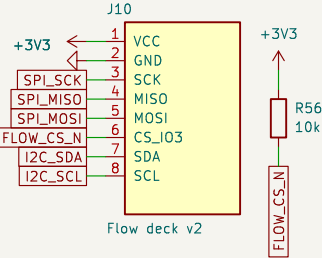
ToF satellite boards (hardware/tof-side x3, hardware/tof-front x1) – wire pads

Side boards: VL53L1X, XSHUT 10 k pull-down on the satellite. Front board: VL53L5CX, LPn/INT/I2C_RST pulls on the satellite. Firmware re-addresses every sensor at boot (all power up at 0x29).



Optical flow – Bitcraze Flow deck v2 (PMW3901 + VL53L1x), wired to pads (OQ-4)

Deck pins used: VCC, GND, SCK, MISO, MOSI, IO_3 (PMW3901 CS), SDA, SCL (Bitcraze deck pinout).
Fed from 3.3 V; the Crazyflie supplies 3.0 V – whether 3.3 V is within the deck's limits is UNCONFIRMED (G2 check).
CS pull-up 10 k keeps the flow sensor deselected during boot.



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN-OHL-S-2.0)

Sheet: /Sensors and expanders/
File: sensors.kicad_sch

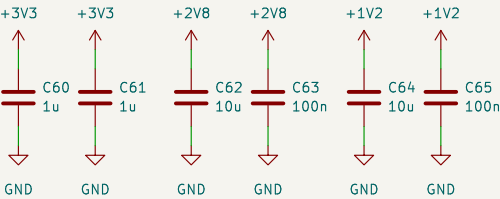
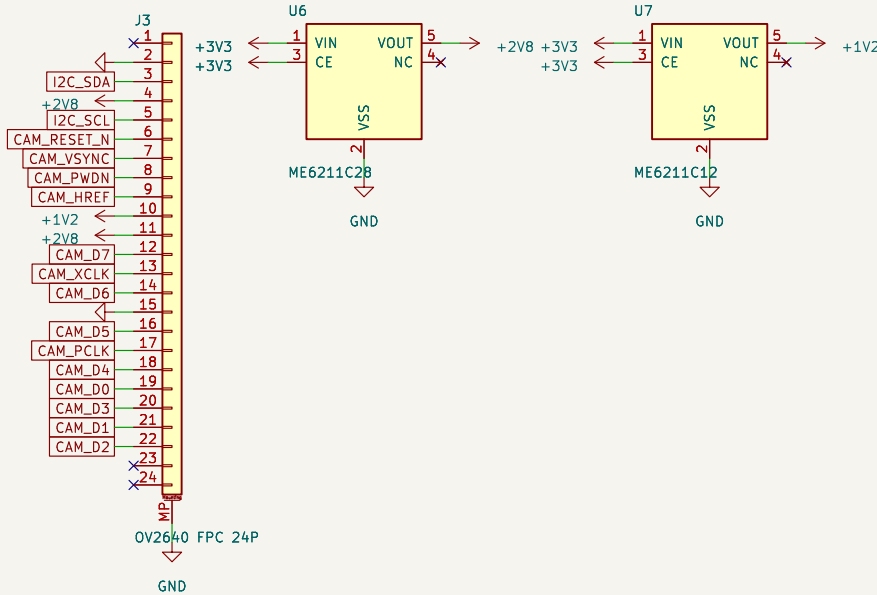
Title: MicroScout flight controller – Sensors and expanders

Size: A3 Date: 2026-10-08 Rev: G2-draft-1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/5

STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

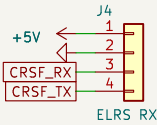
MicroScout flight controller – Camera, LEDs, buzzer, receiver

Camera – OV2640 on a 24-pin 0.5 mm FPC (pinout: Espressif ESP32–S3–EYE MB v2.2, J4)
AVDD + DOVDD 2.8 V (as ESP32–S3–EYE); DVDD 1.2 V (OV2640 datasheet core 1.14–1.26 V; the S3–EYE uses 1.5 V – conflict noted, G2 check against the purchased module). Rails are always on with 3V3 (D–049); PWDN (pulled high) keeps the sensor in standby. SCCB is on the main 3.3 V I2C bus (as on the S3–EYE) – level margin is a G2 check.



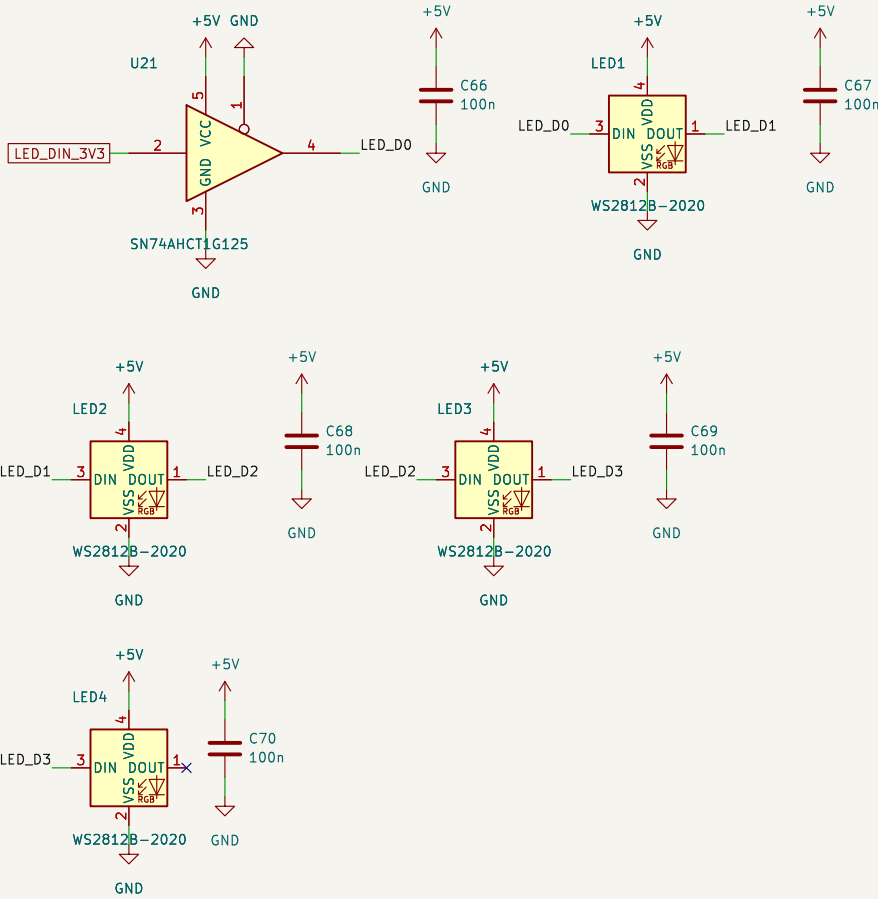
ExpressLRS receiver pads (CRSF, 420 kbaud) – BetaFPV ELRS Lite or similar, powered from 5 V

Receiver TX -> GPIO41 (CRSF_RX); receiver RX <- GPIO42 (CRSF_TX).



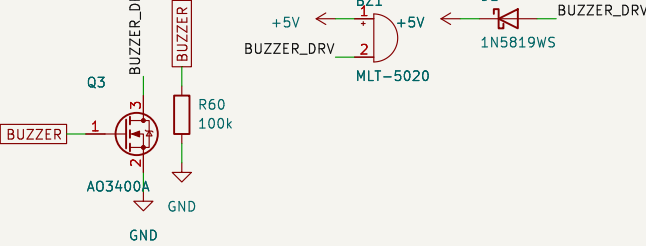
Status LEDs – 4x WS2812B–2020 on 5 V; SN74AHCT1G125 shifts 3.3 V data to 5 V (WS2812B VIH = 0.65 VDD)

Data from GPIO45 via SPI3 MOSI (D–042). 100 nF per LED (agent choice; the LED datasheet says none is needed).



Buzzer – magnetic MLT–5020 driven by AO3400A from 5 V (rated voltage UNCONFIRMED; G2 check)

GPIO46 is a strapping pin that must be low at boot: 100 k gate pull–down keeps it low and the FET off.



Generated schematic – not reviewed. Values are first-pass G2 choices.
STATUS: DRAFT – UNVERIFIED – requires human review at Gate G2

MicroScout (open hardware, CERN–OHL–S–2.0)

Sheet: /Camera, LEDs, buzzer, receiver/
File: io.kicad_sch

Title: MicroScout flight controller – Camera, LEDs, buzzer, receiver

Size: A3 Date: 2026–10–08

Rev: G2–draft–1

KiCad E.D.A. kicad-cli 7.0.11+dfsg–1build4

Id: 5/5